OXFORD AQA
INTERNATIONAL QUALIFICATIONS

INTERNATIONAL GCSE

CHEMISTRY

9202/2

Paper 2

Mark scheme

June 2024

Version: 1.0 Final

246Y9202/2/MS

Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

Further copies of this mark scheme are available from oxfordaqa.com

Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Information to Examiners

1. General

The mark scheme for each question shows:

- the marks available for each part of the question
- the total marks available for the question
- the typical answer or answers which are expected
- extra information to help the Examiner make his or her judgement
- the Assessment Objectives and specification content that each question is intended to cover.

The extra information is aligned to the appropriate answer in the left-hand part of the mark scheme and should only be applied to that item in the mark scheme.

At the beginning of a part of a question a reminder may be given, for example: where consequential marking needs to be considered in a calculation; or the answer may be on the diagram or at a different place on the script.

In general the right-hand side of the mark scheme is there to provide those extra details which confuse the main part of the mark scheme yet may be helpful in ensuring that marking is straightforward and consistent.

2. Emboldening and underlining

2.1 In a list of acceptable answers where more than one mark is available 'any **two** from' is used, with the number of marks emboldened. Each of the following bullet points is a potential mark.

2.2 A bold **and** is used to indicate that both parts of the answer are required to award the mark.

2.3 Alternative answers acceptable for a mark are indicated by the use of **or**. Different terms in the mark scheme are shown by a / ; eg allow smooth / free movement.

2.4 Any wording that is underlined is essential for the marking point to be awarded.

3. Marking points

3.1 Marking of lists

This applies to questions requiring a set number of responses, but for which students have provided extra responses. The general principle to be followed in such a situation is that 'right + wrong = wrong'.

Each error / contradiction negates each correct response. So, if the number of errors / contradictions equals or exceeds the number of marks available for the question, no marks can be awarded.

However, responses considered to be neutral (indicated as * in example 1) are not penalised.

Example 1: What is the pH of an acidic solution?

[1 mark]

Student	Response	Marks awarded
1	green, 5	0
2	red*, 5	1
3	red*, 8	0

Example 2: Name two planets in the solar system.

[2 marks]

Student	Response	Marks awarded
1	Neptune, Mars, Moon	1
2	Neptune, Sun, Mars, Moon	0

3.2 Use of chemical symbols / formulae

If a student writes a chemical symbol / formula instead of a required chemical name, full credit can be given if the symbol / formula is correct and if, in the context of the question, such action is appropriate.

3.3 Marking procedure for calculations

Marks should be awarded for each stage of the calculation completed correctly, as students are instructed to show their working. Full marks can, however, be given for a correct numerical answer, without any working shown.

3.4 Interpretation of 'it'

Answers using the word 'it' should be given credit only if it is clear that the 'it' refers to the correct subject.

3.5 Errors carried forward

Any error in the answers to a structured question should be penalised once only.

Papers should be constructed in such a way that the number of times errors can be carried forward is kept to a minimum. Allowances for errors carried forward are most likely to be restricted to calculation questions and should be shown by the abbreviation ecf in the marking scheme.

3.6 Phonetic spelling

The phonetic spelling of correct scientific terminology should be credited **unless** there is a possible confusion with another technical term.

3.7 Brackets

(…) are used to indicate information which is not essential for the mark to be awarded but is included to help the examiner identify the sense of the answer required.

3.8 Allow

In the mark scheme additional information, ‘allow’ is used to indicate creditworthy alternative answers.

3.9 Ignore

‘Ignore’ is used when the information given is irrelevant to the question or not enough to gain the marking point. Any further correct amplification could gain the marking point.

3.10 Do <u>not</u> accept

‘Do **not** accept’ means that this is a wrong answer which, even if the correct answer is given as well, will still mean that the mark is not awarded.

Question 1

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.1	distillation		1	AO1 3.4.1c

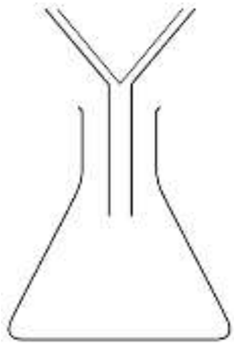
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.2	condenser		1	AO1 3.4.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.3	(liquid) **B** (reason) the liquid has the lowest boiling point	 MP2 is dependent on MP1 being awarded	1 1	AO3 3.4.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.4	systematic error		1	AO4 3.4.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.5	64 (^{o}C)		1	AO2 3.4.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.6	1 ^{o}C		1	AO4 3.4.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
01.7	filter funnel with filter paper inside with a container to collect the filtrate	an answer of  scores **2** marks	1 1	AO4 3.4.1c

Total Question 1		**9**

Question 2

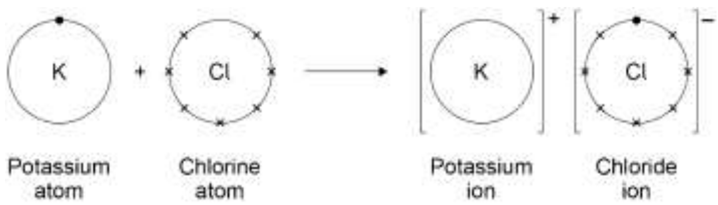
Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.1	halogens		1	AO1 3.2.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.2	both have the same number of outer electrons		1	AO1 3.1.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.3	–101 °C	allow a value in the range –95 °C to –115 °C	1	AO3 3.7.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.4	solid	allow (s)	1	AO3 3.1.1a 3.7.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.5	displacement		1	AO1 3.2.1e 3.7.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.6		allow any combination of x • or $e^{(-)}$ for electrons ignore any inner shells		AO2 3.2.1c,d,e
	(potassium ion) evidence of the outer electron on the potassium atom moving to leave a potassium ion with no electrons in outer shell	allow 8 electrons in outer shell	1	
	(charge) +	allow (charge) +1 / 1+	1	
	(chloride ion) 8 electrons in outer shell		1	
	(charge) –	allow (charge) –1 / 1–	1	
		an answer of  scores **4** marks		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.7	as the surface area increases the percentage reduction in bacteria decreases		1	AO3 3.2.4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
02.8	(surface area to volume = $\frac{15\ 000}{125\ 000}$ =) 0.12		1	AO2 3.2.4
	(**X**) = 0.12:1	allow correct use of an incorrect calculation of surface area to volume	1	

Total Question 2		**12**

Question 3

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.1	(error) start line drawn in ink (improvement) draw start line in pencil (error) water level above the start line (improvement) water below the start line		1 1 1 1	AO3 × 2 AO4 × 2 3.4.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.2	(black ink) any **three** from: • contains (at least) 3 colours • contains a blue colour and a red colour • doesn't contain a green colour • contains one unknown colour	allow (black ink) is a mixture	3	AO3 3.4.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.3	(distance moved by red colour =) 4.8 cm (distance moved by the solvent =) 5.9 cm (R_f =) $\frac{4.8}{5.9}$ = 0.81	allow a value in the range of 4.7 to 4.9 (cm) allow correct use of incorrectly determined distance(s) allow 0.813559 correctly rounded to at least 2 significant figures	1 1 1 1	AO2 × 2 AO3 × 2 3.4.1d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
03.4	(ink) fast violet (reason has the highest R_f value so) travels furthest up the paper		1 1	AO3 3.4.1e

Total Question 3		**13**

Question 4

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.1	universal indicator (solution / paper) (pH =) 3	allow pH meter / probe allow a value in the range 1–6 ignore any references to colour	1 1	AO1 3.5.1f 3.10.3.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.2	does not ionise completely (when dissolved) in water	allow does not ionise completely (when dissolved) in aqueous solution	1	AO1 3.10.3.2b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.3	H_2O	allow H–O–H	1	AO1 3.10.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.4	ester		1	AO3 3.10.3.3a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.5	any **two** from: • volatile • distinctive / pleasant smells • not toxic / corrosive		2	AO1 3.10.3.3b

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
04.6	$\left(\text{moles of ethanoic acid} = \frac{1.5}{60}\right)$ = 0.025 (mol)		1	AO2 3.6.1d
	(0.025 moles ethanoic acid = 0.025 moles ethyl ethanoate) (mass ethyl ethanoate =) 0.025 × 88	allow correct use of incorrectly determined moles of ethanoic acid	1	
	= 2.2 (g)		1	
	alternative approach 60 g ethanoic acid produces 88g ethyl ethanoate (1) 1.5 g ethanoic acid produces = $\frac{88}{60}$×1.5 (g) ethyl ethanoate (1) = 2.2 (g) (1)			

Total Question 4		**10**

Question 5

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.1	sulfur is produced (which is a) solid	 allow (which is a) precipitate	1 1	AO2 3.5.2c 3.8.1a

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.2	$\frac{31 + \mathbf{X} + 38}{3} = 34$ **X** = 102 – 69 = 33 (s)	 allow **X** = 102 – 31 – 38	1 1 1	AO2 3.8.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.3	there is very little spread of repeat values around the mean	allow the results (for 8 g/dm^3) are close to the mean allow there is a narrow range of results (for 8 g/dm^3)	1	AO4 3.8.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.4	*y*-axis scale (0) 20, 40, 60, 80	ignore intermediate values	1	AO2 × 3 AO3 × 1 3.8.1e
	all points correctly plotted	allow a tolerance of ± ½ a small square allow 1 mark for 2 or 3 points correct	2	
	line of best fit		1	

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.5	as concentration (of sodium thiosulfate solution) increases the time taken (for the cross to be no longer visible) decreases		1	AO3 3.8.1e

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
05.6	(as concentration increases) the rate of reaction increases		1	AO1 3.8.1e
	(because) there are more particles per unit volume	allow (because) the particles are closer together	1	
	(so) there is a higher frequency of collisions		1	

Total Question 5		**14**

Question 6

Question	Answers	Mark	AO/ Spec. Ref.
06.1	**Level 3**: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO1 × 4 AO3 × 2 3.4.3b 3.4.3c 3.4.3e
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • add sodium hydroxide solution • copper ions form a blue precipitate • iron(II) ions form a green precipitate • both calcium and magnesium ions form white precipitates • add silver nitrate solution • (in the presence of) nitric acid solution • chloride ions form a white precipitate • bromide ions form a cream precipitate • this allows calcium chloride and magnesium bromide to be distinguished to access Level 3, the expected results of the tests should be included		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
06.2	potassium sulfate	allow K_2SO_4 allow a compound of potassium for **1** mark allow a compound of sulfate for **1** mark	2	AO2 3.4.3a,f

Total Question 6		**8**

Question 7

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.1	7		1	AO1 3.5.1f

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.2	sodium sulfate		1	AO2 3.5.1c

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.3	$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$	allow for 1 mark $H^+ + OH^- \rightarrow H_2O$	2	AO1 3.5.1g

Question	**Answers**	**Mark**	**AO/ Spec. Ref.**
07.4	**Level 3**: The method would lead to the production of a valid outcome. The key steps are identified and logically sequenced.	5–6	AO4 3.6.4b
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content: • **measure 25.0 cm^3 of sodium hydroxide solution into a conical flask** • measure using a pipette • **fill the burette with sulfuric acid** • place conical flask on a white tile • **add a few drops of indicator to the conical flask** • **add sulfuric acid** • in increments of 1 cm^3 • swirl the flask • add the sulfuric acid dropwise • **until indicator changes colour** • **record the volume of sulfuric acid added** • repeat method until two results within 0.1 cm^3 are achieved		

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
07.5	(conversion 28.10 cm^3 sulfuric acid =) 0.02810 dm^3		1	AO2 3.6.4c
	(moles sulfuric acid = 0.02810 × 0.20 =) 0.00562 mol	allow correct use of incorrect / no conversion of volume	1	
	(moles sodium hydroxide = 0.00562 × 2 =) 0.01124 mol	allow correct use of incorrectly determined moles of sulfuric acid	1	
	$\left(\text{concentration} = \frac{0.01124}{0.0250}\right)$ = 0.4496 (mol/dm^3)	allow correct use of incorrectly determined moles of sodium hydroxide	1	
	= 0.450 (mol/dm^3)	allow a correctly rounded answer to 3 significant figures from an incorrect calculation which uses the data in the question	1	
	alternative approach:			
	$\left(\text{ratio } \frac{\text{moles } H_2SO_4}{\text{moles NaOH}} =\right)$	allow inverted expression		
	$\frac{1}{2} = \frac{28.1 \times 0.20}{25.0 \times \text{concentration}}$ (2)	allow **1** mark for the expression with an incorrect mole ratio		
	concentration = $\frac{28.1 \times 0.20 \times 2}{25.0}$ (1)	allow correct use of the expression with an incorrect mole ratio		
	=0.4496 (1)			
	=0.450 (1)	allow a correctly rounded answer to 3 significant figures from an incorrect calculation which uses the data in the question		

Total Question 7		**15**

Question 8

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.1	(ethanol) is made from a renewable raw material (ethanol) has a lower carbon footprint **or** (ethanol) contributes less to climate change		1 1	AO3 3.10.1.2g

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.2	$C_2H_5OH + 2\,O_2 \rightarrow 2\,\mathbf{CO} + 3\,H_2O$	allow **1** mark for CO irrespective of balancing number allow for **2** marks $C_2H_5OH + 2\,O_2 \rightarrow \mathbf{C + CO_2} + 3\,H_2O$	2	AO2 3.10.1.2d

Question	Answers	Extra information	Mark	AO/ Spec. Ref.
08.3	(mass ethanol in solution = 0.250×21.0 =) 5.25 (g)		1	AO2 3.6.3a 3.6.3b 3.6.4a
	(moles of ethanol =) $\frac{5.25}{46}$	allow correct use of an incorrectly determined mass of ethanol	1	
	= 0.1141 (mol)		1	
	(number of molecules =) $0.1141 \times 6.02 \times 10^{23}$	allow correct use of an incorrectly determined number of moles of ethanol	1	
	$= 6.87 \times 10^{22}$		1	

Total Question 8		**9**